CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, CAD/CAM)

January 2014

ME – 701.01 Computer Aided Design

Date: 06.01.2014, Monday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator & PSG data book is allowed.

SECTION- I

Q1 (a) Differentiate between classical design & computer aided design process. 5

(b) Answer following with correct option. 6

1. Which of the following is not graphics standard:
   (A) GKS (C) UNIX
   (B) IGES (D) PHIGS
2. C^2 continuity refers to:
   (A) Common tangent (C) Common curvature
   (B) Common point (D) All of the above
3. Cube is represented by
   (A) 12 vertices and 18 edges (C) 12 vertices and 12 edges
   (B) 8 vertices and 8 edges (D) 8 vertices and 12 edges
4. Sweep can be generated by following method
   (A) Linearly (C) Both Linearly and Non - linearly
   (B) Non - linearly (D) Exponentially
5. Tablet cursors is
   (A) Digitizer (C) Light pen
   (B) Joystick (D) Programmable dials
6. A convenient method of representing geometrical transformations is by introducing the
   (A) Cartesian coordinate system (C) Polar coordinate system
   (B) Homogeneous coordinate system (D) All of the above

Q2 (a) Compare the capabilities of types of displays. 3

(b) Explain the working principle of digitizer. 3

(c) Using Bresenham's line algorithm, find the pixel positions between end points (10, 5) and (15, 9). 6

OR

Q2 (a) What is scan conversion? Why can't a vector scan display produce as many colours as a raster scan display? 3

(b) Differentiate between Analytic curves and Parametric curves. 3

(c) How can you generate a circle with centre 'c' and radius 'r' using Bresenham's algorithm. 6

Q3	(a)	Compare geometric transformation with coordinate transformation. Also explain composite transformation.	3
	(b)	What is homogenous coordinate system? Why is it preferred in computer graphics?	3
	(c)	A square having end points A(1,1), B(6,1), C(6,6), D(1,6) is rotated by 50° clockwise direction keeping point (6,1) fixed. Find the final coordinates.	6

OR

Q3	(a)	Write 3-D transformation matrix for rotation about x, y, z-axis.	3
	(b)	Prove that 3D rotations are not commutative.	3
	(c)	Find the reflection of point (5, 5) about a line $y = 3x + 8$.	6

SECTION- II

Q4	(a)	Write down the steps for generation of solid model of a Hexagonal Bolt using any solid modeling software.	4
	(b)	Explain parametric continuity conditions for curves.	4
	(c)	Explain most common mating conditions for locating and orienting parts in their assembly.	3
Q5	(a)	Explain in brief different type of scanners? What are active scanner and passive scanner?	6
	(b)	A cubic Bezier curve is defined by control points (1, 2) (3, 4) (6, -6) & (10, 8). Calculate the coordinates of parametric point of the curve at u = 0.2, 0.4, 0.6, 0.8.	6

OR

Q5	(a)	Describe the concept of parametric programming and feature based modeling.	4
	(b)	By giving suitable examples explain the various techniques used for surface modeling.	4
	(c)	Calculate the parametric midpoint of Hermite cubic curve that fits the points $P_0 = (1, 3)$, $P_1 = (7, 2)$ and the tangent vectors $P'_0 = (10, 8)$, $P'_1 = (6, 0)$.	4
Q6	(a)	Write a short notes on the following: 1) DXF - Format 2) STL – Format	6
	(b)	Write an interactive program and algorithm for design of helical compression springs.	6

OR

Q6	(a)	Discuss the need for standardization in computer graphics & list the various CAD standards.	6
	(b)	Prepare an exhaustive algorithm for design of spur gear.	6

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